DSP Exercise Set: Signals & Time-Domain Analysis

Theory Questions

1. Define the difference between energy signals and power signals. Give one real-world example of each.

Energy signals must have a finite energy (strictly greater than 0), and a zero average power, while power signals have an infinite energy, and a finite average power (strictly greater than 0).

For example, a Tae-Kwon-Do kick serves as an energy signal, while the transmission of the radio station called *Alfa 91.3* serves as a power signal.

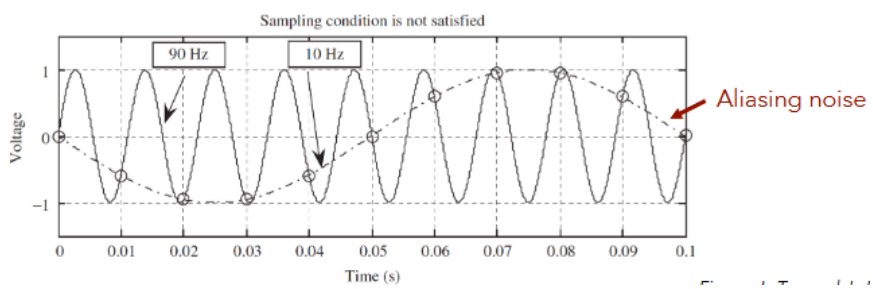
2. State the Nyquist sampling criterion and explain what happens if it is not satisfied.

$$f_s \geq 2f_{\max}$$

Where:

- f_s is the sampling frequency and
- $f_{\max}$ is the maximum frequency of the evaluated signal

If the Nyquist sampling criterion isn't satisfied, then our sampled signal won't catch the natural behavior of the continuous signal. An example of this phenomenon can be found in the figure below.



3. Explain in your own words how time reversal and time shifting affect a signal's graph. Why are these operations important in signal analysis?

I would define **time reversal** as a mirror reflex of the signal, where the mirror is the y-axis. On the other hand, I would define **time shifting** as the horizontal displacement of a signal.

These operations are important because we use them to process our digital signals, to get a defined result. I think that if we know or study as many operations as we can, we will improve our capacity of designing systems that can lead us to reach those defined objectives,

Exercises

1. For each of the following signals, determine whether it is continuous/discrete, periodic/aperiodic, and causal/non-causal:

a) x(t) = sin(2π10t) -> Continuous – Periodic – Non-causal.
b) x[n] = u[n-3] -> Discrete – Aperiodic - Causal
c) x(t) = e^{-t}u(t) -> Continuous – Aperiodic – Causal

2. Is the signal x[n] = cos((5π/6)n) periodic? If yes, find its fundamental period.

A discrete-time signal x[n] is periodic, with period N, if and only if

$$x[n+N] = x[n] \quad \forall \quad \infty \leq n \leq \infty$$

In our case, a cosine signal like cos (Ωn) requires the following condition:

$$\frac{\Omega}{2\pi} = \frac{m}{N}$$

Where:

- Ω is the angular frequency and
- m and N must be integer numbers.

Then, we know that:

$$\Omega = \frac{5\pi}{6}$$

Replacing:

$$\frac{\frac{5\pi}{6}}{2\pi} = \frac{m}{N}$$

Reducing:

$$\frac{5}{12} = \frac{m}{N}$$

Thus, m and N are integer numbers, and $\boldsymbol{N = 12}$ **is the fundamental period** that we were looking for.

3. **A continuous-time signal x(t) = cos(200πt) is sampled at fs = 150 Hz.**
   - **What is the Nyquist rate?**

The Nyquist rate is equivalent to the sampling frequency that satisfies the following condition:

$$f_s = 2f_{\max}$$

The general form of a cosine signal is:

$$cos(2\pi f t)$$

We're analyzing a signal that occurs in a single frequency, so:

$$f = f_{max}$$

Thus, the Nyquist rate of our signal is:

$$f_s = \frac{2(200\pi)}{2\pi} = 200[Hz]$$

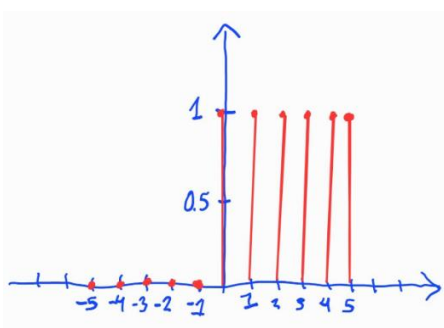
- Will aliasing occur? If so, what is the apparent frequency after sampling?

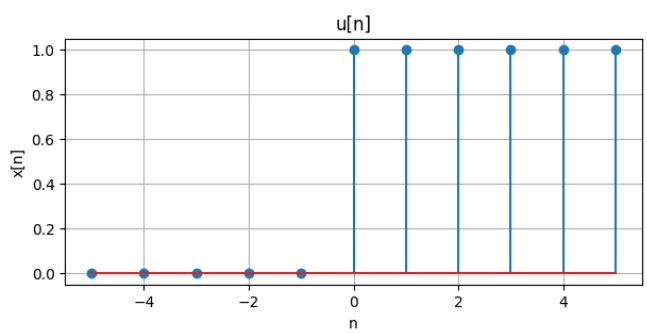
Given that the f_s established is lower than the Nyquist rate, it will occur aliasing.

To obtain the apparent frequency after sampling, we need to calculate the subtraction between f_s and f_0:

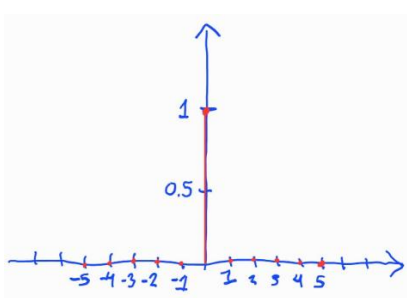
$$f_a = |f_s - f_0| = |150 - 100| = 50\ [Hz]$$

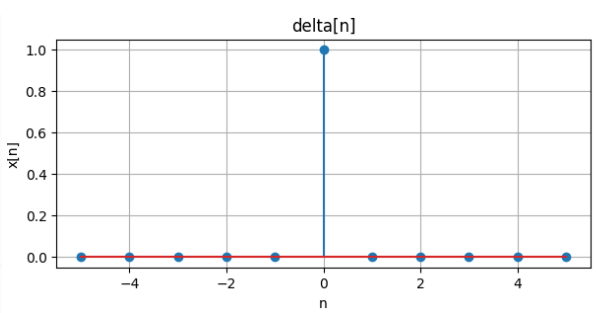
4. **Sketch the following signals for -5 ≤ n ≤ 5:**
   a) u[n]



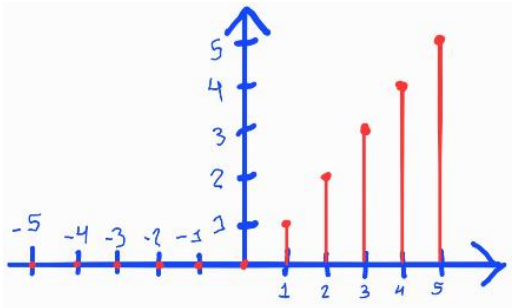


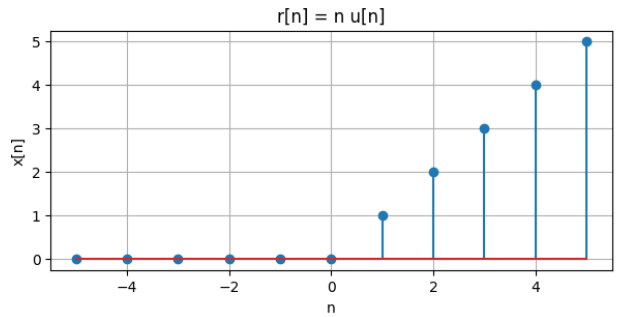
b) δ[n]



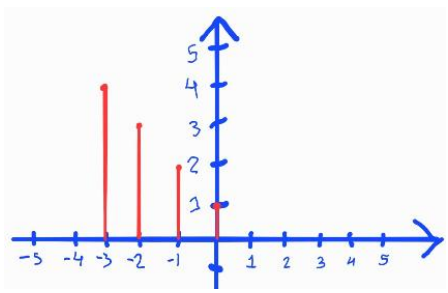


c) r[n] = n u[n]





5. Given x[n] = {1,2,3,4} for n=0,1,2,3, compute and sketch x[-n].



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[4. 3. 2. 1.]
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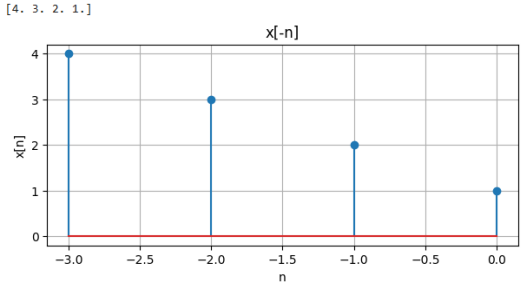
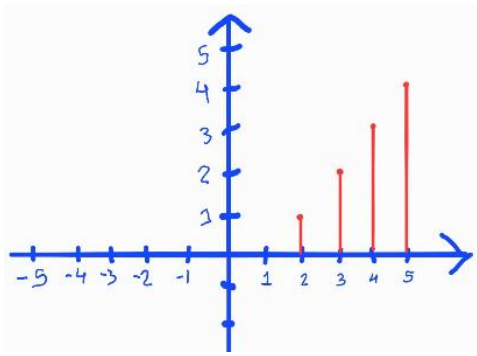


6. For the same signal x[n] = {1,2,3,4}, compute and sketch:

a) x[n-2]



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[0. 0. 1. 2. 3. 4.]
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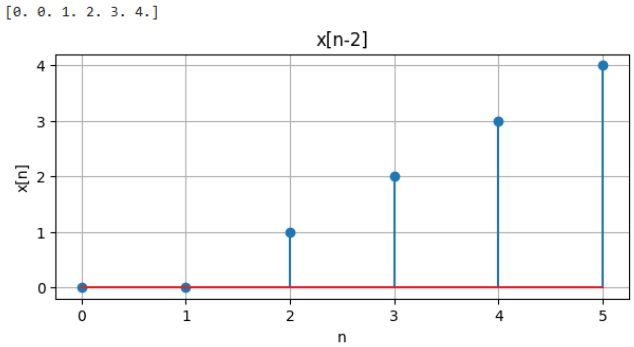
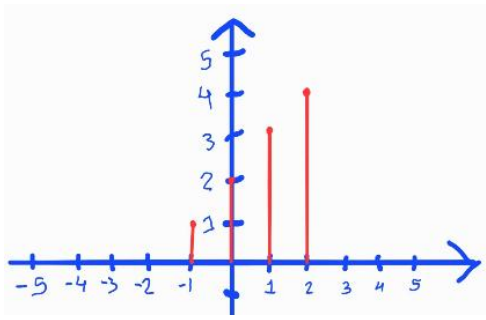


b) x[n+1]



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[1. 2. 3. 4. 0. 0.]
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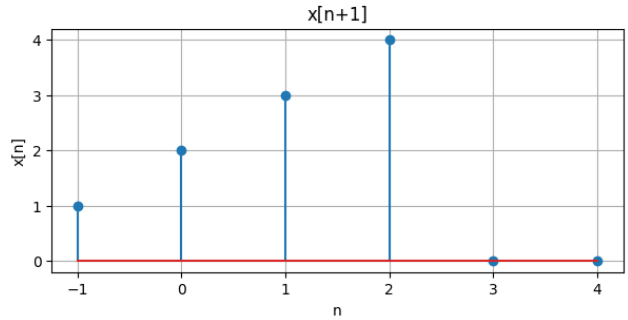
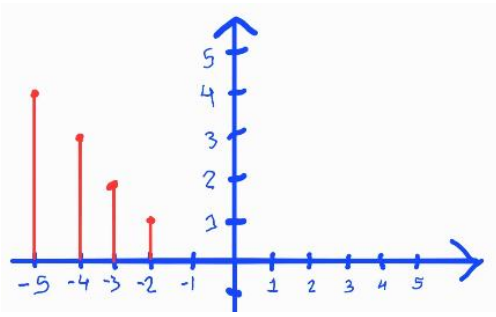


7. Take x[n] = {1,2,3,4}. Compute and sketch:

a) x[-n+2]



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[4. 3. 2. 1. 0. 0.]
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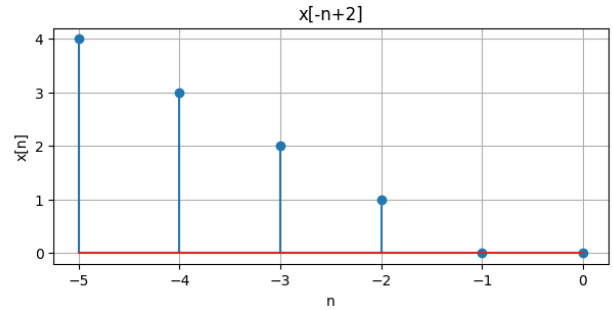
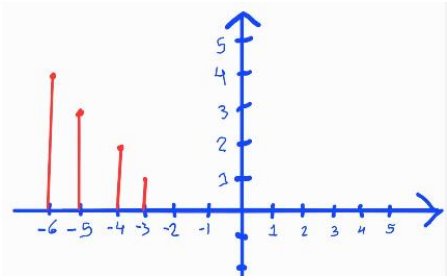


b) x[3-n]



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[4. 3. 2. 1. 0. 0. 0.]
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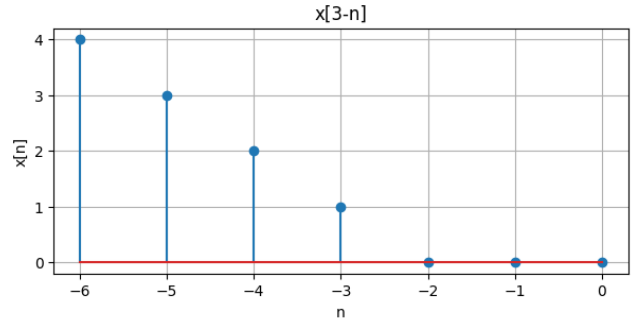
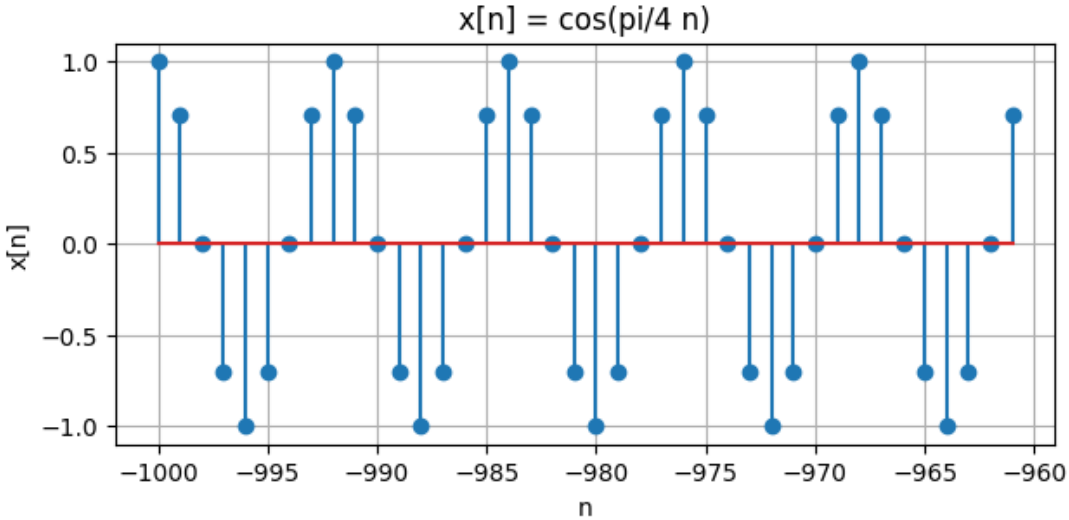


8. Determine whether the following signals are energy signals or power signals:

a) $x[n] = \cos((\pi/4)n)$ power signal

b) $x[n] = (1/2)^n u[n]$ energy signal

Compute energy/power as appropriate.

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Potencia aproximada: 0.5000000000000002
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Potencia aproximada: 0.5000000000000002
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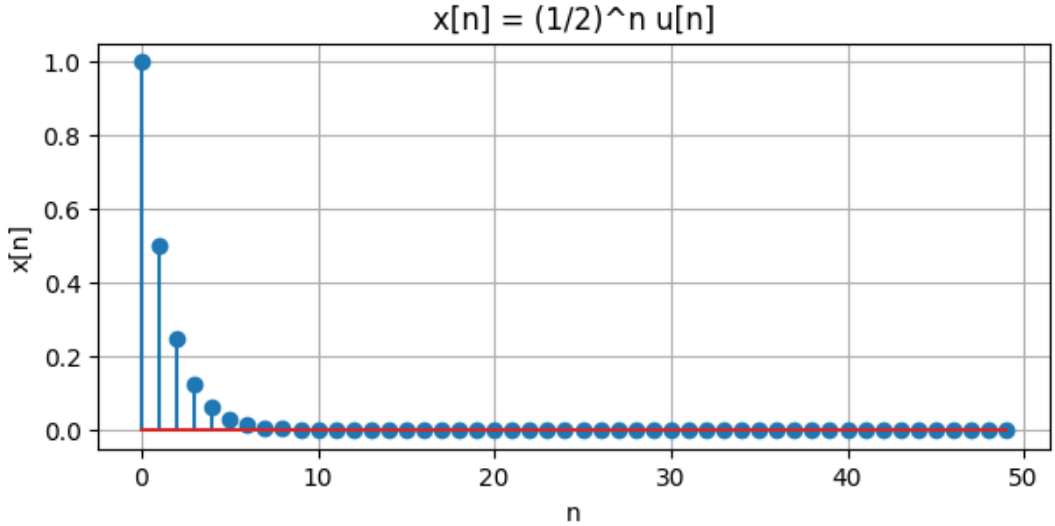


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Energía aproximada: 1.3333333333333333
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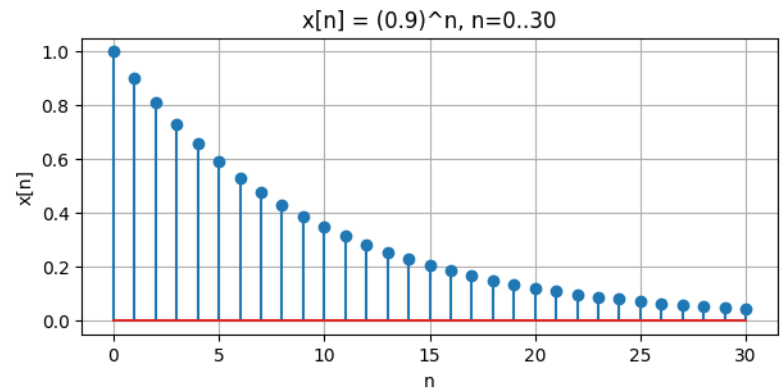
9. Given x[n] = $(0.9)^n$ u[n]:

- Is this an energy or power signal? Energy signal
- Compute its total energy.

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Energía (truncada 0..30): 5.255496956089839
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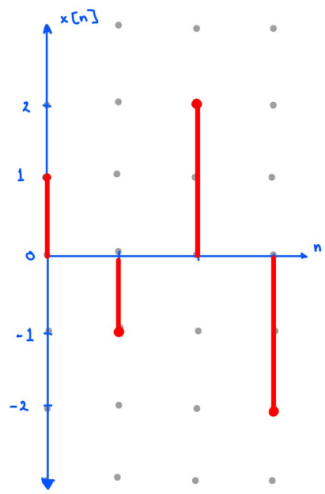
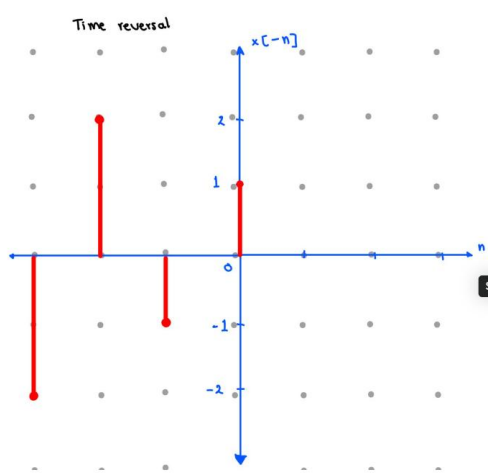
10. A signal x[n] = {1,-1,2,-2} for n=0,1,2,3.

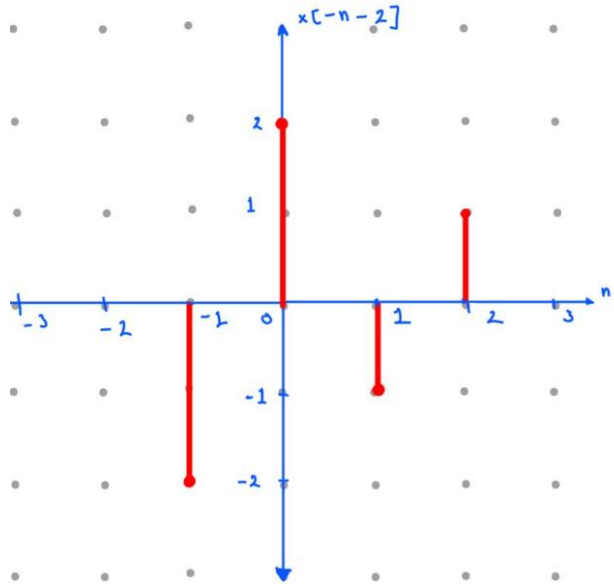
- Plot the original signal.
- Apply time reversal → shift right by 2 samples → scale by factor 3
- Write the resulting signal explicitly and sketch it.



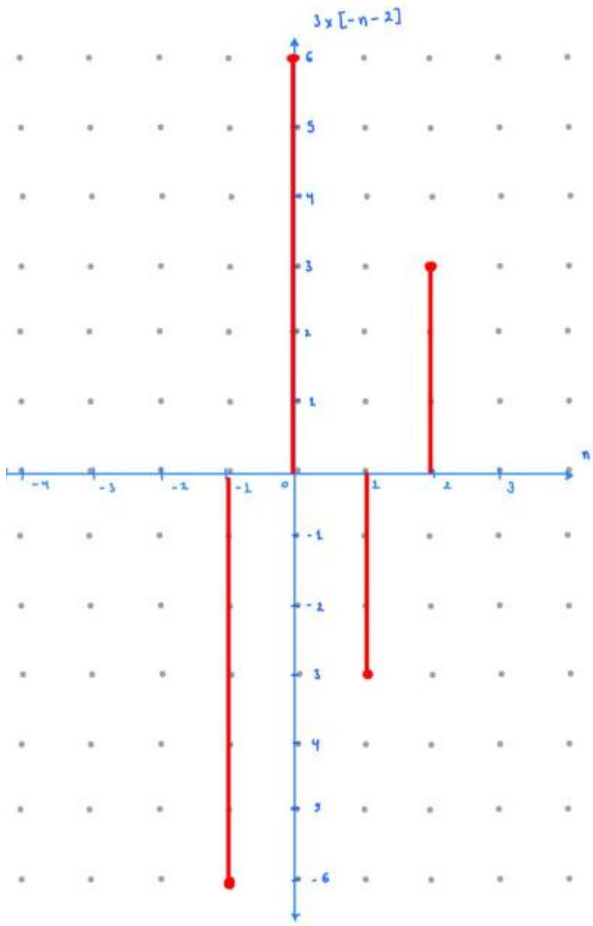
Apply time reversal: x[-n] = {-2,2,-1,1} for n= -3, -2, -1, 0



Shift right by 2 samples: x[-n-2] = {-2,2,-1,1} for n= -1, -0, 1, 2



Scale by factor 3: 3x[-n-2] = {-6,6,-3,3} for n= -1, -0, 1, 2



11. Compute the mean and RMS of x[n] = {2, -2, 2, -2} for n=0...3.

$$RMS = \sqrt{\frac{(2)^2 + (-2)^2 + (2) + (-2)^2}{4}}$$

$$RMS = 2.0$$

12. Compute the autocorrelation of x[n] = {1,2,1} and cross-correlation with y[n] = {1,0,-1}.

$x[n] = \{1, 2, -1\}$

Autocorrelation

$k = -2$

$R_{xx} = (1)(-1) = -1$

$k = -1$

$R_{xx} = (1)(2) + (2)(-1) = 0$

$k = 0$

$R_{xx} = (1)(1) + (2)(2) + (-1)(-1) = 6$

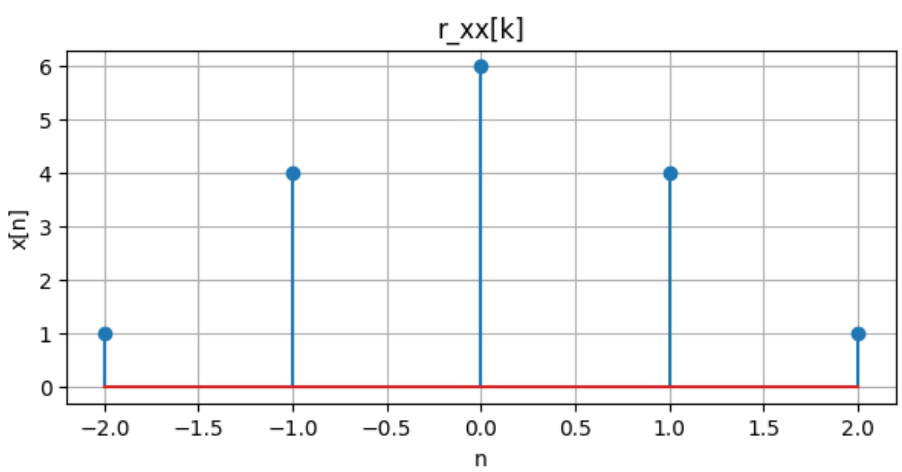
$k = 1$

$R_{xx} = (2)(1) + (-1)(2) = 0$

$k = 2$

$R_{xx} = (-1)(1) = -1$

...



Cross-correlation

$x[n] = \{1, 2, 1\}$

$y[n] = \{1, 0, -1\}$

$K = -2$

$R_{xy} = (1)(-1) = -1$

$K = -1$

$R_{xy} = (1)(0) + (2)(-1) = -2$

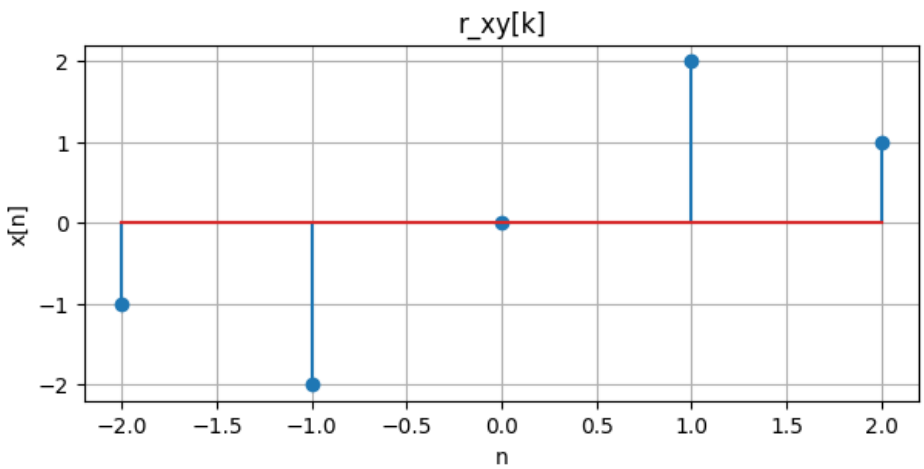
$K = 0$

$R_{xy} = (1)(1) + (2)(0) + (1)(-1) = 0$

$K = 1$

$R_{xy} = (2)(1) + (1)(0) = 2$

$K = 2$

$R_{xy} = (1)(1) = 1$

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